

Data-Driven Analysis of Repeat Purchase and Customer Churn Using Large-Scale Retail data

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I. INTRODUCTION

The fashion apparel retail industry is highly lucrative, dynamic, and unpredictable, as it is shaped by a wide range of factors including cultural influences, economic conditions, social media exposure, and rapidly changing fashion trends. Because retailers must continuously innovate, track evolving customer preferences, and respond to global trends while frequently introducing new products, demand and sales forecasting in this sector is particularly challenging. The high level of uncertainty in fashion retail makes forecasting methods commonly used in other industries more difficult to apply effectively [1].

From a business perspective, one-time buyers are often considered high-cost, low-value customers. If they are not clearly identified, their presence can distort key performance metrics and lead to inefficient marketing decisions [2].

In addition, customer churn is a critical issue in the retail sector, as acquiring new customers is typically more expensive than retaining existing ones. High churn rates therefore reduce long-term profitability by increasing marketing and acquisition costs [3].

At the same time, modern retail environments generate large volumes of customer data across multiple channels. Machine learning and data-driven approaches allow retailers to transform these data into personalised marketing strategies that improve customer engagement, strengthen brand loyalty, and increase conversion rates [7]. As a result, marketing analytics supported by machine learning has become an essential tool for effective retail management.

Against this background, this study focuses on the concepts of repeat purchase and customer churn using data-driven analysis and machine learning techniques. Customers are divided into one-time buyers and multi-day buyers, with particular attention given to understanding the characteristics of both buyers. The study aims to support repeat purchase conversion, reduce churn risk, and contribute to retail management strategies by proposing insights based on both customer behaviour and product characteristics.

II. ANALYTICAL QUESTIONS

* The notebook code follows the order of the questions below.

1. Sampling large-scale retail data

How can a representative sample be extracted from a large-scale retail transaction dataset while preserving key statistical properties of the full population?

2. Timing of repeat purchases

When do repeat purchases most often occur after a customer's first purchase, and which factors increase the likelihood of repeat purchase?

3. Behavioural segmentation of one-day and repeat customers

Do one-day and repeat buyers form distinct behavioural clusters, and how can these differences inform targeted marketing strategies?

4. Churn risk among repeat customers

Among customers who have made at least one repeat purchase, which customers are associated with a higher risk of churn?

5. Product preference differences between one-day and heavy users

Do one-day buyers and heavy users differ in the types and characteristics of products purchased, and can these differences be measured using product similarity analysis?

III. DATA

A. Key Characteristics

The original dataset consists of three datasets: **Transactions**, **Articles**, and **Customers**.

The transaction data includes the keys 'customer_id' and 'article_id', which are shared with the customer and article datasets. Using these keys, the three datasets can be merged with the transaction data as the main data.

	About	Features	Observations
Transaction	Transactional data	7	31,788,324
Articles	Product info	25	105,542
Customer	Customer info	4	1,371,980

The transaction dataset is large (3.49GB), so sampling was applied to enable efficient processing. After preprocessing, the final dataset contains 32 features and 534,666 observations. Additional feature engineering was conducted to better capture customer behaviour, resulting in a behavioural dataset with 16 features and 332,804 observations.

B. Suitability of the data

- After merging the three datasets, the processed data will include both product and customer information, allowing analysis at the customer and product levels.
- Date variables will be used to define repeat purchase and churn. By aggregating transactions by customer ID and date, customers will be grouped into one-day and multi-day buyers.
- The product description variable 'detail_desc' will be used to compare product characteristics and similarities.

IV. ANALYSIS

A. Data preparation

1) Sampling

We loaded over 3GB of transaction data using the Dask library and applied two different sampling methods. The first method divided the full dataset into 54 partitions and sampled 1,000 rows from each partition, resulting in a total of 54,000 observations. The second method randomly sampled 1% of the full dataset.

To evaluate how well each sample represented the original data, we compared the mean and standard deviation of numerical variables between the population and the samples, and compared the value proportions of categorical variables. Based on these comparisons, we selected the first sample.

Dimension	Metric	Population	S1	S2
Data size	Row Count	31,788,320	540,000	317,880
Price	Mean	0.027829	0.027779	0.027906
	Std	0.019181	0.019240	0.019283
Product distribution	Top 1 Product Share	0.158%	0.157%	0.169%
	Top 2 Product Share	0.110%	0.108%	0.101%
Temporal Distribution	Peak Date Share	0.624%	0.625%	0.624%
K-S test			0.00199	0.00253

2) Dealing with Missing Data

Only the variable 'detail_desc' contains missing values. Since this variable is not used in questions 1–4 of the analysis, the missing values were left unchanged.

3) Creating behavioral Data through Feature Engineering

To analyse customer behaviour and build churn classification and clustering models, a separate behavioural dataset was created.

This dataset includes engineered features that describe different aspects of customer behaviour, such as recency and frequency of purchases, spending behaviour, product diversity, temporal and sales channel patterns, and basic customer attributes.

B. Survival analysis

This study applies survival analysis to examine when customer repurchase is most likely to occur. Survival analysis

is a statistical method used to model the time until a specific event happens, such as customer churn or repeat purchase.

The event refers to the outcome of interest and is commonly defined as a point where a state change occurs, such as death, disease onset, system failure, or customer churn.

Since this dataset does not contain an explicit churn indicator (0/1), the event is defined as the occurrence of a second purchase, and the duration is measured as the time from the first purchase to the second purchase. Customers who do not make a second purchase by the end of the observation period are treated as right-censored, indicating that the event has not yet occurred.[3]

Status	Interpretation	Event
No repeat purchase yet	Customer is in the survival state	0
Repeat purchase occurs	Event has occurred	1

$$S(t)=P(T>t)$$

The survival function is defined as the **probability that a repurchase has not occurred by time t**

The steep decline in the early period supports the suitability of survival analysis for capturing early repurchase dynamics.

The Cox proportional hazards model is widely used to analyse how predictor variables influence survival time, and is applied here to examine the effect of first purchase spend on repeat purchase timing. [4]

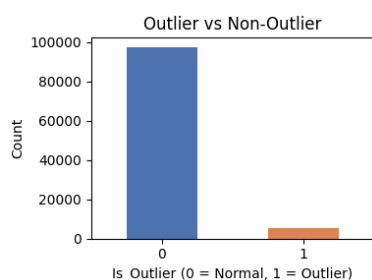
C. Modelling - Clustering

Customers were separated into one-day buyers and multi-day buyers, and clustering was performed within each group to support more detailed and targeted marketing strategies.

Muli-day-buyer : Customers who purchased more than one day

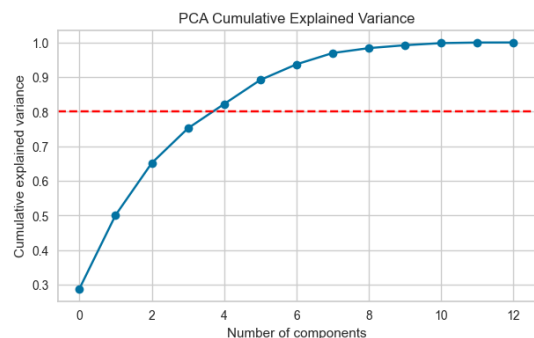
1) Outlier

To improve clustering quality, outliers were considered for removal. However, as extreme values in this dataset may represent high-value customers, we used a conservative approach and only the top **5% of outliers** were removed.



2) PCA

PCA was used prior to clustering to simplify the data and highlight the main patterns driving customer behaviour. The number of PCA components was set to **five**, corresponding to the point where the cumulative explained variance reaches approximately 80%, ensuring that most of the information in the data is retained while reducing dimensionality.



3) Select K

Based on the elbow point, the number of clusters was set to **three**.

4) Clustering Models

Cluster Distribution by Method	
Method	Cluster Distribution
K-Means	- Cluster 0: 52.36% - Cluster 2: 30.57% - Cluster 1: 17.07%
GMM	- Cluster 0: 52.36% - Cluster 2: 30.57% - Cluster 1: 17.07% (Same as K-Means)
DBSCAN	- Cluster 2: 19.71% - Cluster 0: 16.32% - Cluster 4: 15.13% - Noise (-1): 7.66% - Remaining clusters < 6% (42 clusters total)

- **K-Means** was used to group data points into distinct clusters revealing patterns efficiently. [5]
- **GMM** was applied to allow probabilistic cluster membership. [5]
- **DBSCAN** was used to detect density-based clusters and identify noise customers. [5]
- **Agglomerative (hierarchical) clustering** was not applied due to high computational cost.

Interestingly, K-means and GMM produced identical cluster proportions, indicating that the cluster structure in the PCA space is close to spherical and well separated. This suggests a stable and robust segmentation.

Each category represented over 10% of the sample, allowing them to be included in the analysis.

From a business perspective, **K-means was chosen** because it provides clear and easily interpretable customer segments

and delivers stable results that can be readily translated into actionable marketing strategies.

One-day-buyer : Customers who purchased only one day within the observation period

One-day buyers show highly dispersed behaviour, resulting in weak and poorly defined clusters with little separation by behavioural intensity.

D. Modelling - Classification

1) Target

Churn is defined based on previous RFM studies, where customers with long inactivity are considered churned.[8] In this study, informed by the survival analysis, customers who did not repurchase within 180 days by the end of the observation period are classified as churned.

(target distribution = 1 : 1)

2) IV (Information Value) checking & Optimal binning

Those were applied to improve interpretability, but it did not substantially improve model performance. Therefore, they were used for explanatory purposes.

3) Encoding

Logistic regression uses one-hot encoding for categorical variables, while tree-based models apply label encoding.

4) These three models were selected to compare a stable and interpretable baseline (LR) with tree-based models of increasing complexity (RFC and LGBM), in order to evaluate the trade-off between model simplicity, overfitting risk, and performance potential.

Model Performance Comparison

Model	F1 (Train)	F1 (Test)	AUC (Train)	AUC (Test)
LR	0.6578	0.6583	0.6274	0.6226
RFC	0.9856	0.5694	0.9993	0.5872
RFC (BO)	0.6457	0.6471	0.6283	0.6242
LGBM	0.6632	0.6448	0.6774	0.6290

E. Product preference differences analysis (NLP)

1) This study compares products bought by one-day and multi-day buyers using NLP-based cosine similarity. Product descriptions are turned into text vectors with Sentence-BERT, allowing us to measure how similar the products are. Because cosine similarity focuses on direction rather than size [5], it provides a stable way to compare products and helps design targeted recommendations to encourage repeat purchases.

2) Used model: all-MiniLM-L6-v2

all-MiniLM-L6-v2 is a lightweight sentence encoder that converts text into semantic vectors for tasks like similarity, clustering, and retrieval. [9]

```

similarity_matrix
✓ 0.0s
array([[0.45012605, 0.4416058 , 0.64404094, ..., 0.49297166, 0.54001766,
        0.36983734],
       [0.6162558 , 0.4457583 , 0.40459383, ..., 0.6840872 , 0.44824246,
        0.42620093],
       [0.70490885, 0.6661702 , 0.49553522, ..., 0.5916183 , 0.38356715,
        0.8541805 ]],

```

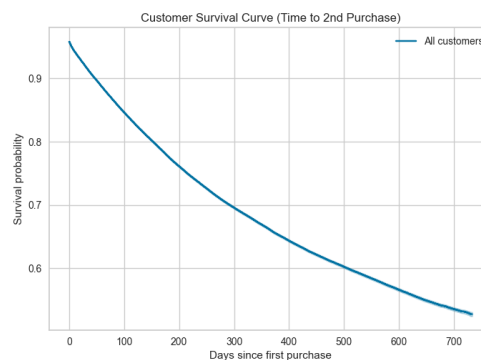
3) Used indicators:

- The maximum similarity: The closest match between a one-day buyer product and heavy-user products
- The mean similarity: How closely the product aligns with heavy-user preferences overall.

V. FINDINGS

1. Timing of repeat purchases

1) Kaplan–Meier curve



The largest decline in the survival curve occurs within the first 0–200 days, indicating that repeat purchases occur most frequently during this period and that the hazard of repurchase is highest. Therefore, we define customers who do not make a repeat purchase within 180 days as churned, as their likelihood of future repurchase is substantially reduced.

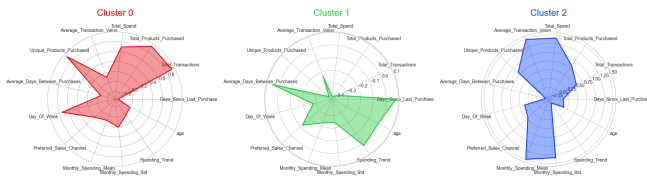
2) Cox PH model

- Effect of first purchase spend on repeat purchases

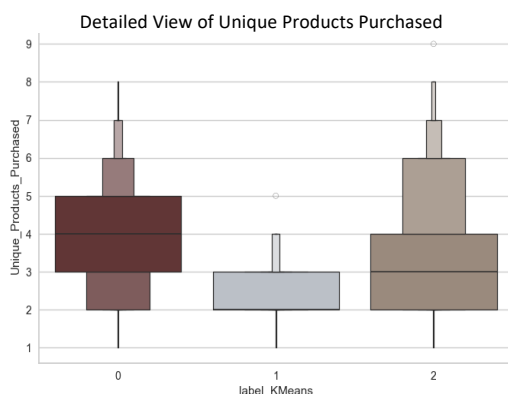
	coef	exp(coef)	se(coef)
log_first_purchase_spend	0.29	1.33	0.00
Concordance		0.57	
Partial AIC		2725729.84	
log-likelihood ratio test		3850.61 on 1 df	
-log2(p) of ll-ratio test		inf	

The exponentiated coefficients are interpretable as multiplicative effects on the hazard.[4] Exp(coef), (a hazard ratio) of 1.33 implies that increasing first purchase spend by roughly 2.7 times is associated with a 33% higher likelihood of repeat purchase, indicating that higher initial spend can meaningfully improve customer retention.

2. Behavioural segmentation of one-day and repeat customers



0	Customers who purchase frequently but with low average order value, and mainly shop on weekends. → Use bundle deals and add-on recommendations to increase basket size during weekend campaigns.
1	Older customers with long purchase intervals, low transaction amounts, and the highest risk of churn. → Focus on reactivation offers and reminder-based campaigns with simple, low-commitment incentives
2	Younger customers with short purchase intervals, high transaction values, and diverse style preferences. → Apply personalised recommendations and early-access promotions to maintain engagement and loyalty.

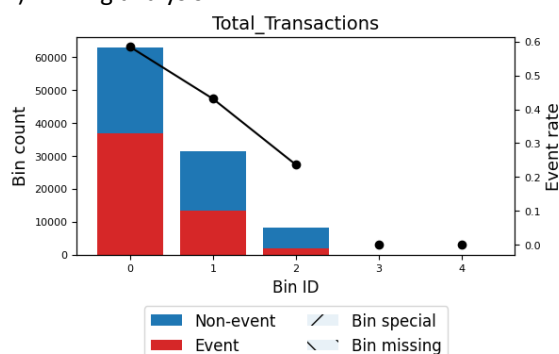


3. Churn risk among repeat customers

1) Model selection

Logistic Regression is selected as it shows less evidence of overfitting and achieves the best balance between precision and recall among the evaluated models. However, given the 1:1 target class ratio, further improvements should be achieved through additional model tuning, refined feature selection, and clearer target definition or adjustment.

2) Binning analysis



Since churn remains high for customers with up to two purchases (binning0) but decreases sharply from four

purchases onward (binning2), retention efforts should focus on accelerating early repeat purchases. Targeted promotions such as fourth-purchase incentives, progress-based loyalty programs, and early-stage re-engagement campaigns can help move customers beyond this critical threshold and improve long-term retention.

4. Product preference differences between one-day and heavy users

1) Comparing similarity

Low similarity product

max_sim_to_heavy	mean_sim_to_heavy
Moisturising lip balm in a plastic pot with...	Metal studs in various designs.
Moisturising lip balm in a pot shaped like...	Moisturising lip balm in a pot shaped like...
Moisturising lip balm with a watermelon flavour	Moisturising lip balm in a rabbit-shaped plastic
.....	

When similarity is high, one-day buyers and heavy buyers purchase very similar products. This suggests the issue is **not the product itself**, but timing or engagement. These products are strong **transfer products** and are good candidates for **repurchase reminders**, **post-purchase recommendations**, and **same-category expansion**.

High similarity product

max_sim_to_heavy	mean_sim_to_heavy
Scarf in a soft, textured knit with...	Long-sleeved, fitted top in soft, organic cotton...
Calf-length dress in soft jersey crêpe with..	Jacket in an airy, crinkled viscose weave with...
Vest top in cotton jersey with an inset mesh...	Knee-length shirt jacket in soft faux
.....	

When similarity is low, one-day buyers purchase products that are quite different from what heavy buyers prefer. This indicates **limited conversion potential**. These items are more likely to be **entry-level**, **event-driven**, or **purpose-specific purchases**. Instead of aggressive retention, **short-term promotions and seasonal campaigns** are more effective.

2) Hidden Differences in Similar Products

Product	max	mean
Sparkly stones Alice band	0.327	0.126
Glittery horns Alice band	0.331	0.165

Even within the same product category, semantic similarity scores reveal differences in alignment with heavy-user preferences, highlighting candidates for prioritised recommendations.

VI. REFLECTIONS AND FUTURE WORK

1) Although we expected to find multiple patterns among one-time buyers, the high proportion of them limited the ability to form meaningful customer clusters resulting in weak clustering outcomes.

During the machine learning phase, model performance was influenced by the limited availability of strongly predictive features. This emphasised the importance of maintaining alignment between research objectives, data characteristics, and analytical approaches.

Market basket analysis was also explored to identify cross-selling opportunities based on co-purchased items. However, same-day multi-item purchases by individual customers were rare, which limited the feasibility of this approach.

2) Future research could improve churn prediction by incorporating additional data sources and applying further model tuning. Using product image data may also enable a more detailed comparison of preferences between one-time and repeat buyers. In addition, recommendation strategies based on frequently purchased items within customer clusters could be explored.

VII. WORD COUNTS / REFERENCE

Introduction	286/300
Analysis Questions / Data	296/300
Analysis	952/1000
Findings, Reflections and Further Work	597/600

[1] M. Koren, N. Abulafia, L. Shviro and N. Weiss, (2024) "A Machine Learning Approach to Forecasting Demand in Fashion Industry," 2024 4th International Conference on Advanced Enterprise Information System (AEIS) pp. 52-56

[2] Zhao, B., Yahyapour, R., Takasu, A. and Fu, X. (2019) *Loyal consumers or one-time deal hunters: Repeat buyer prediction for e-commerce*. In: Proceedings of the 2019 International Conference on Data Mining Workshops (ICDMW). IEEE, pp. 1080–1087.

[3] Larivière, B. and Van den Poel, D. (2004) Investigating the role of product features in preventing customer churn, by using survival analysis and choice modeling: The case of financial services. *Expert Systems with Applications*, 27(2), pp. 277–285.

[4] Fox, J. and Weisberg, S. (2011) *Cox Proportional-Hazards Regression for Survival Data in R*. Appendix to *An R Companion to Applied Regression*, 2nd

[5] Araújo, T. (2023) *Informed retail product similarity*. PhD thesis, Universidade do Porto, Portugal. ProQuest Dissertations & Theses Global.

[6] Patil, A.P., M P, D., Mittal, S., Shetty, S., Hiremath, S.S. and Patil, Y.E. (2017) Customer churn prediction for retail business. Proceedings of the International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS 2017). Bangalore, India: IEEE, pp. 845–850.

[7] Islam, M.R., Shawon, R.E.R. and Sumsuzoha, M. (2023) Personalized marketing strategies in the U.S. retail industry: Leveraging machine learning for better customer engagement. *International Journal of Machine Learning Research in Cybersecurity and Artificial Intelligence*, 14(1)

[8] Granov, A. (2021) Customer loyalty, return and churn prediction through machine learning methods. Master's thesis. Umeå: Umeå University, Department of Mathematics and Mathematical Statistics.

[9]<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

Code References

- Customer behaviour feature engineering and radar chart visualization

<https://www.kaggle.com/code/farzadnekouei/customer-segmentation-recommendation-system>

- Clustering visualization (boxenplot)

<https://www.kaggle.com/code/karnikakapoor/customer-segmentation-clustering>

- Binning

FastCampus online machine learning course, Chapter 29

Declaration

- ChatGPT was used to assist in writing code to sample transaction data using partitioning.

- ChatGPT was used to support graph design, including colour selection and line styles.

- ChatGPT was used to generate the “*Distribution of Max Similarity to Heavy Users*” graph.

- ChatGPT was used to generate tables summarising clustering model outcomes in this report.

- ChatGPT was used to review and correct grammatical errors